



Inflammation

Lec. 4

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Inflammation

- Inflammation is a response of vascularized tissues to infections and tissue damage that brings cells and molecules of host defense from the circulation to the sites where they are needed, to eliminate the offending agents.
- Inflammation is a beneficial host response to foreign invaders and necrotic tissue, but also may cause tissue damage.

Inflammatory conditions are termed by adding the **suffix (-itis)** to the affected organ or tissue.

For example:-

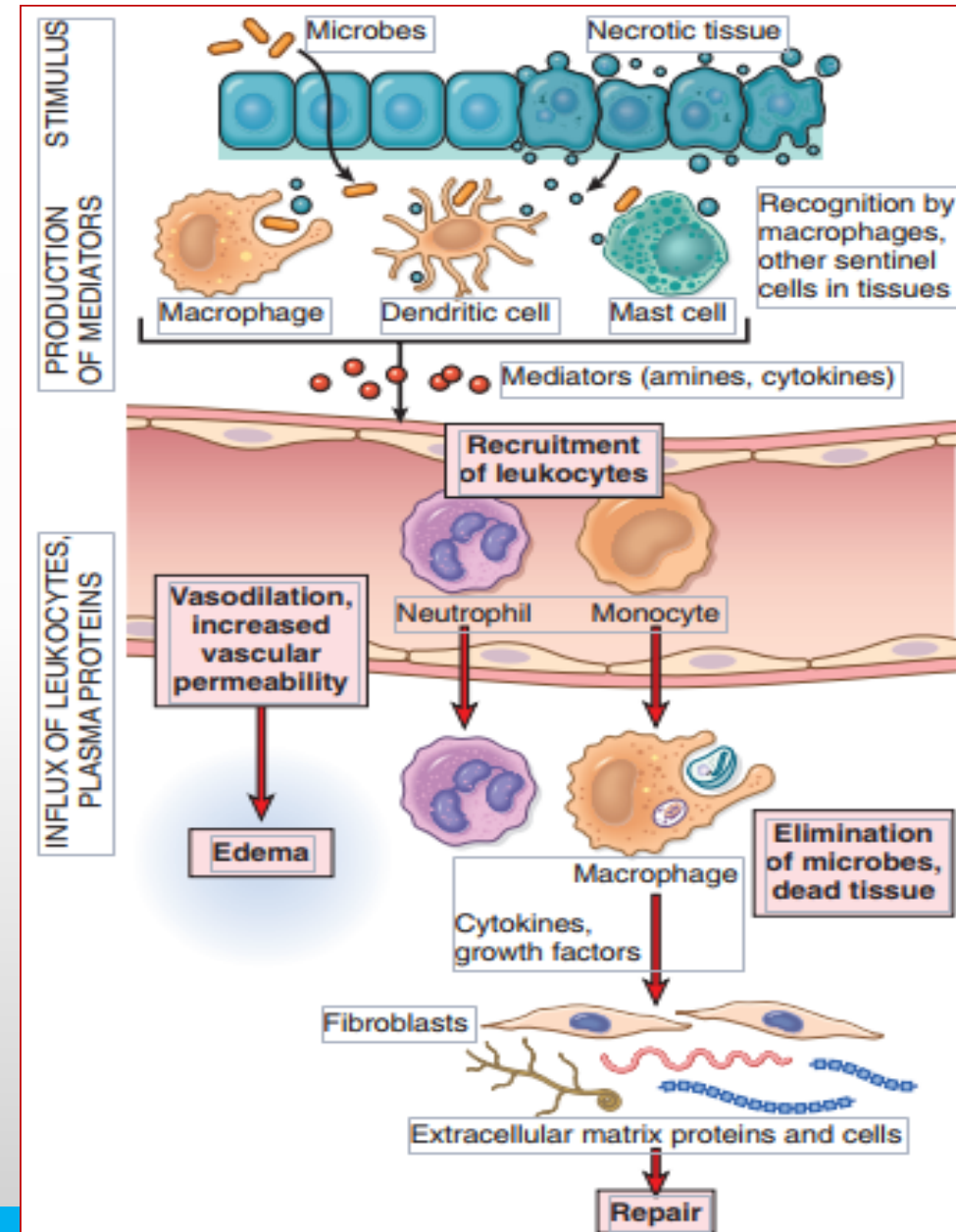
Pulpitis : inflammation of pulp

Gingivitis: inflammation of gingiva

Hepatitis: inflammation of liver

Appendicitis: inflammation of appendix

- Inflammation is induced by chemical mediators that are produced by host cells in response to injurious stimuli. When a microbe enters a tissue or the tissue is injured, the presence of the infection or damage is sensed by resident cells, including macrophages, dendritic cells, mast cells, and other cell types.



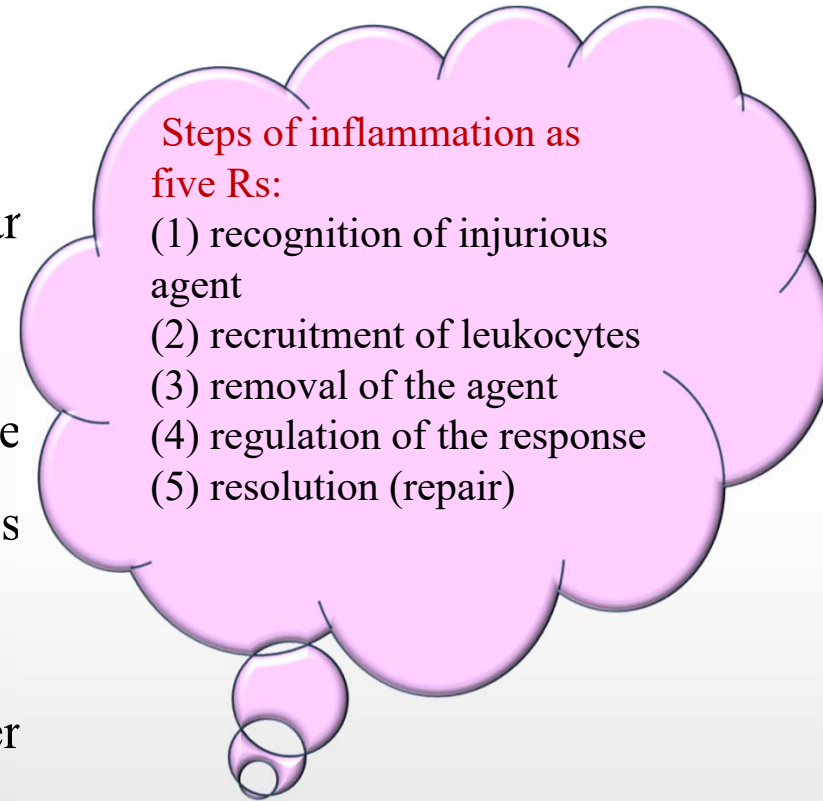
- These cells secrete molecules (cytokines and other mediators) that induce and regulate the subsequent inflammatory response. Inflammatory mediators are also produced from plasma proteins that react to the microbes or to products of necrotic cells. Some of these mediators promote the efflux of plasma and the recruitment of circulating leukocytes to the site where the offending agent is located. Mediators also activate the recruited leukocytes, enhancing their ability to destroy and remove the offending agent.

The typical inflammatory reaction develops through a series of sequential steps:

- The offending agent, which is located in extravascular tissues, is recognized by host cells and molecules.
- Leukocytes and plasma proteins are recruited from the circulation to the site where the offending agent is located.
- The leukocytes and proteins are activated and work together to destroy and eliminate the offending substance.
- The reaction is controlled and terminated.
- The damaged tissue is repaired.

Steps of inflammation as five Rs:

- (1) recognition of injurious agent
- (2) recruitment of leukocytes
- (3) removal of the agent
- (4) regulation of the response
- (5) resolution (repair)



Causes of inflammation

- **Infections** (bacterial, viral, fungal, parasitic) and microbial toxins
- **Tissue necrosis** elicits inflammation regardless of the cause of cell death, which may include ischemia , trauma, and physical and chemical injury .
- **Foreign bodies** (splinters, dirt, sutures) .
- **Immune reactions** (also called hypersensitivity) against environmental substances or against self tissues.

Types of inflammation

1- Acute inflammation

2-Chronic inflammation

Feature	Acute	Chronic
Onset	Fast: minutes or hours	Slow: days
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury, fibrosis	Usually mild and self-limited	May be severe and progressive
Local and systemic signs	Prominent	Less

Cardinal signs of inflammation

The external manifestations of inflammation, often called its cardinal signs, are :

- 1-Redness
- 2- Swelling
- 3-Heat
- 4-Pain
- 5-loss of function

❖ *These signs are typically more prominent in acute inflammation than in chronic inflammation .*



Acute inflammation

- Acute inflammation: initial, rapid response to infections and tissue damage. It typically develops within minutes or hours and is of short duration, lasting for several hours or a few days. Its main characteristics are the exudation of fluid and plasma proteins (edema) and the emigration of leukocytes, predominantly neutrophils (also called polymorphonuclear leukocytes).

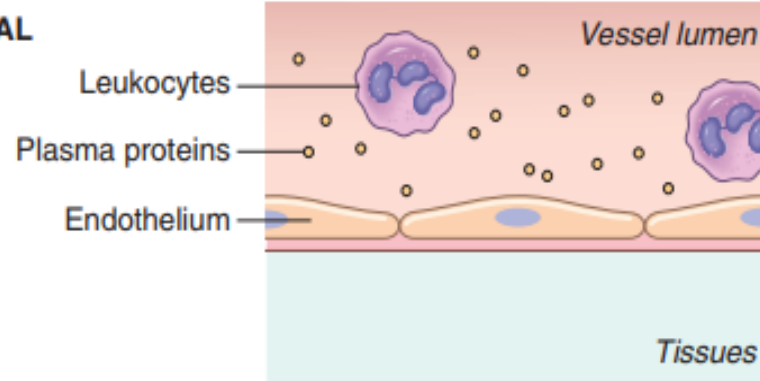
- Acute inflammation has two major components :

- 1) Vascular Changes

- 2) Cellular Changes

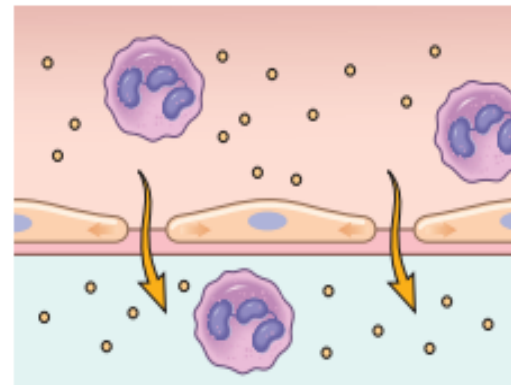
- **Vascular changes** : designed to maximize the movement of plasma proteins and leukocytes out of the circulation and into the site of infection or injury.
 - a) **Changes in vascular flow and caliber** : Initial transient vasoconstriction, followed by vasodilation causing increasing in blood flow to the area of injury.
 - b) **Increased vascular permeability (vascular leakage)** : leading to the escape of a protein-rich fluid (exudate) into the extravascular tissue results in edema. The mechanisms of increased vascular permeability are **retraction of endothelial cells** leading to formation of endothelial gap and **endothelial injury as by burn**.

A NORMAL



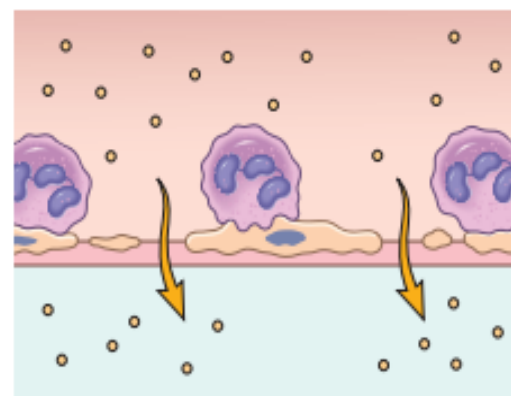
B RETRACTION OF ENDOTHELIAL CELLS

- Induced by histamine, other mediators
- Rapid and short-lived (minutes)

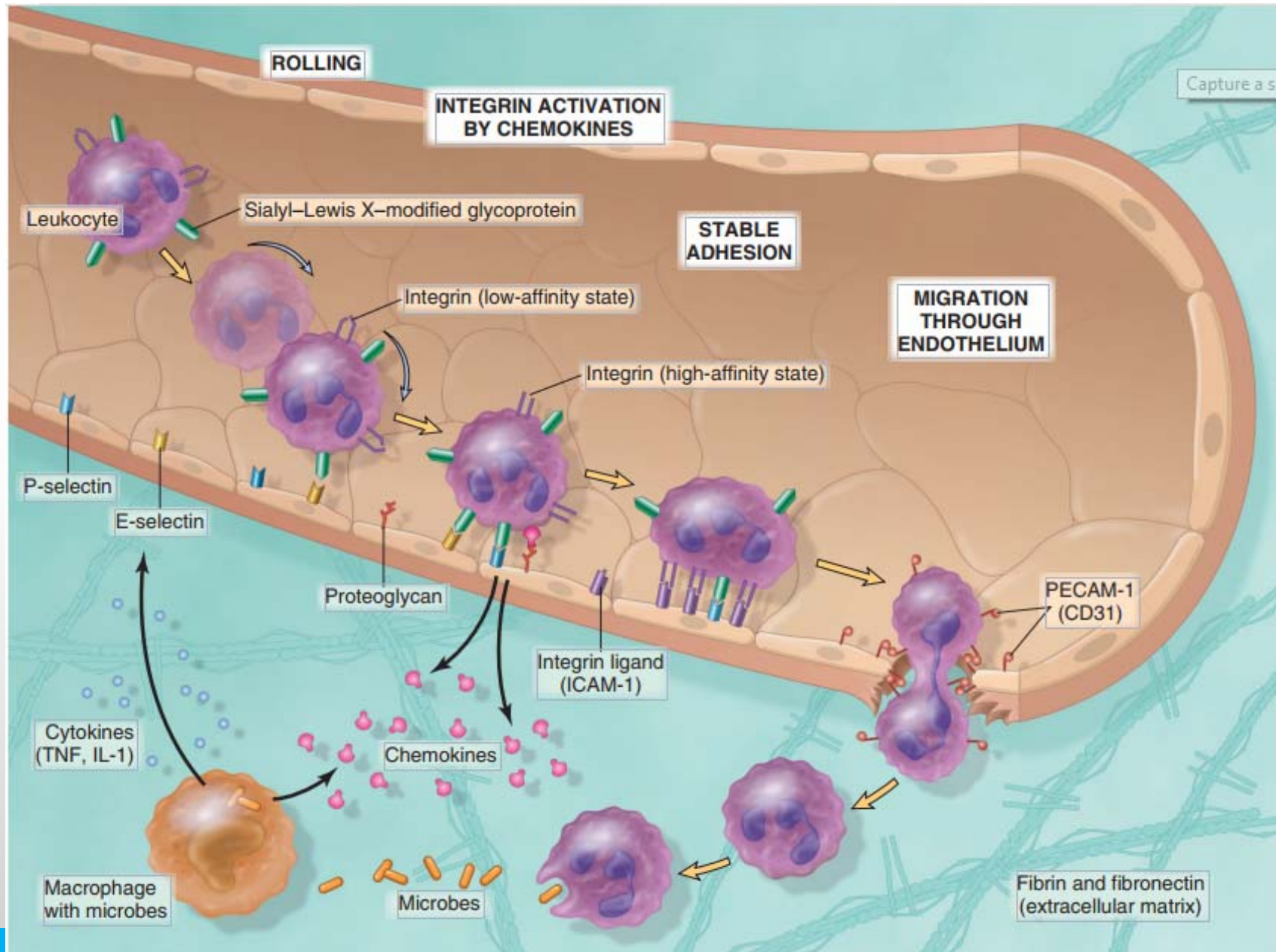


C ENDOTHELIAL INJURY

- Caused by burns, some microbial toxins
- Rapid; may be long-lived (hours to days)



- **Cellular Changes** : Leukocytes are recruited from the blood into the extravascular tissue where infectious pathogens or damaged tissues may be located, migrate to the site of infection or tissue injury, and are activated to perform their functions.
- ***Leukocyte recruitment is a multistep process consisting of :***
 - 1-Margination and rolling
 - 2-Adhesion
 - 3- Migration through interendothelial gaps.
 4. Chemotaxis : Movement of leukocytes toward the inflammatory stimulus in the direction of the locally produced chemoattractants.

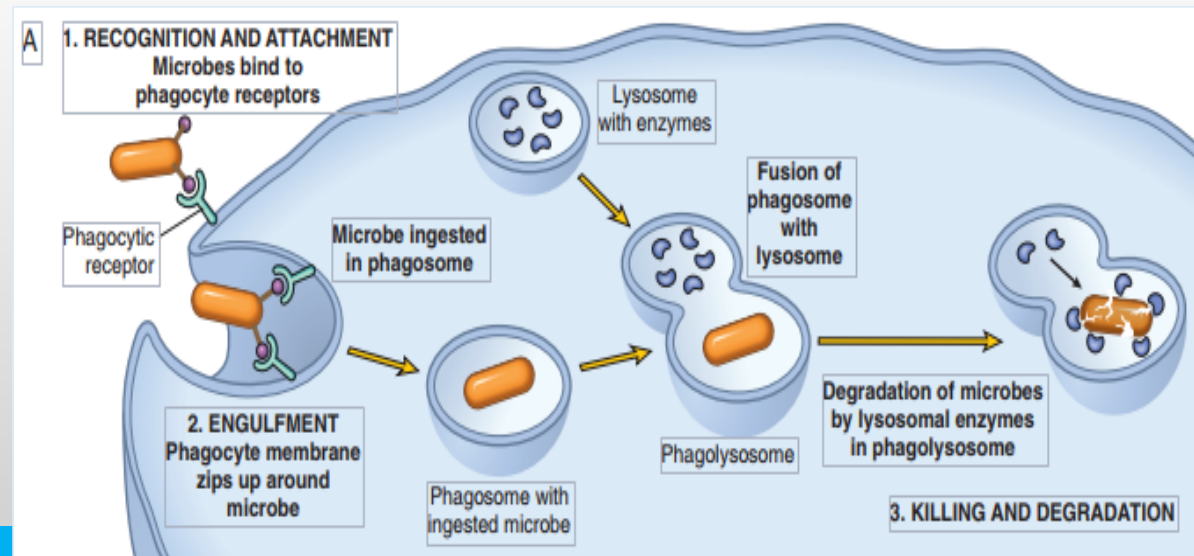


Capture a sp

- **Leukocyte Activation** : Once leukocytes (particularly neutrophils and monocytes) have been recruited to the site of infection or tissue necrosis, they must be activated to perform their functions. Stimuli for activation include microbes, products of necrotic cells, and several mediators .
- The functional responses that are most important for destruction of microbes and other offenders are phagocytosis and intracellular killing.

Phagocytosis involves three sequential steps:

1. Recognition and attachment of the particle to be ingested by the leukocyte
2. Engulfment, with subsequent formation of a phagocytic vacuole
3. Killing or degradation of the ingested material



Outcomes of acute inflammation

- 1) Resolution : regaining normal histological & functional tissue.
- 2) Progression to chronic inflammation
- 3) Suppuration : means pus formation
- 4) Scarring and fibrosis : it means replacement of the injured tissue by fibrous connective tissue

Thank You